

Root systems

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Chapter 1

Root pairings

1.1 Definitions

Recall that a *perfect pairing* is a quadruple $(R, M, N, \mathcal{L})$ where R is a ring, M and N are R -modules, and $\mathcal{L} : M \times N \rightarrow R$ is a bilinear map such that the induced maps $M \rightarrow \text{Hom}_R(N, R)$ and $N \rightarrow \text{Hom}_R(M, R)$ are isomorphisms of R -modules. If I is a set, denote by $\text{Perm}(I)$ the group of permutations of I .

Definition 1. A *root pairing* is a tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ where $(R, M, N, \mathcal{L})$ is a perfect pairing, I is a set, $\alpha : I \rightarrow M$, $i \mapsto \alpha_i$, $\alpha^\vee : I \rightarrow N$, $i \mapsto \alpha_i^\vee$, and $s : I \rightarrow \text{Perm}(I)$, $i \mapsto s_i$, satisfying:

1. α and $\alpha^\vee$ are injective,
2. for all $i \in I$, $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$,
3. for all $i, j \in I$,

$$\begin{aligned}\alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee)\alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee)\alpha_i^\vee.\end{aligned}$$

Definition 2. Fix a root pairing $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$. We say that Φ is:

- *crystallographic* if $\mathcal{L}(\alpha_i, \alpha_j^\vee) \in \mathbb{Z}$ for all $i, j \in I$,
- *reduced* if, for all $i, j \in I$, the roots α_i and α_j are linearly dependent over R if and only if $\alpha_i = \pm\alpha_j$,
- *finite* if I is finite.

Chapter 2

Constructions

2.1 Type A

Consider the perfect pairing

$$(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, \langle \cdot, \cdot \rangle),$$

where $\langle \cdot, \cdot \rangle$ is the canonical pairing between $\mathbb{Z}^n$ and its algebraic dual $(\mathbb{Z}^n)^* = \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{Z}^n$, and let $e_1^*, \dots, e_n^*$ be the dual basis.

Write $[1, n] = \{1, \dots, n\}$. An *interval* in $[1, n]$ is a set $[i, j] = \{i, i+1, \dots, j\}$ with $1 \leq i \leq j \leq n$. We denote by I_n the set of intervals in $[1, n]$ and write $[i] = [i, i]$.

A *signed interval* in $[1, n]$ is an element of $I_n \times \{\pm 1\}$. We identify $J \in I_n$ with $(J, 1)$ and write $-J$ for $(J, -1)$.

Let $A = \mathcal{A}_n = (a_{u,r})$ be the $n \times n$ integer matrix defined by

$$a_{u,r} = \begin{cases} 2 & \text{if } u = r, \\ -1 & \text{if } |u - r| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

This is the Cartan matrix of type A_n .

Define

$$\alpha : I_n \times \{\pm 1\} \rightarrow \mathbb{Z}^n, \quad \alpha_{\pm J} = \pm \sum_{r \in J} A e_r, \quad (2.1)$$

$$\alpha^\vee : I_n \times \{\pm 1\} \rightarrow (\mathbb{Z}^n)^*, \quad \alpha_{\pm J}^\vee = \pm \sum_{r \in J} e_r^*. \quad (2.2)$$

Lemma 3. *Let $J = \varepsilon[i, j]$ and $K = \delta[k, l]$ be signed intervals. Then*

$$\mathcal{L}(\alpha_K, \alpha_J^\vee) = \varepsilon \delta(\delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{k,j+1}).$$

Proof. It is enough to treat positive intervals, since the signs factor out by bilinearity. For fixed r , the vector $A e_r$ has coefficient 2 in coordinate r , coefficient -1 in the adjacent coordinates $r-1$ and $r+1$, and coefficient 0 elsewhere. Hence, for $u \in [1, n]$,

$$\sum_{r=k}^l a_{u,r} = \delta_{u,k} + \delta_{u,l} - \delta_{u,k-1} - \delta_{u,l+1},$$

where $\delta_{x,y}$ is the Kronecker delta. Summing over $u = i, \dots, j$ gives the displayed formula. $\square$

Lemma 4. For every $J \in I_n \times \{\pm 1\}$, $\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2$.

Proof. Write $J = \varepsilon[i, j]$. In Lemma 3, the two positive endpoint terms are both 1, while the two negative endpoint conditions $i = j + 1$ are impossible because $i \leq j$. Thus

$$\mathcal{L}(\alpha_J, \alpha_J^\vee) = \varepsilon^2(1 + 1) = 2.$$

□

Lemma 5. For signed intervals J and K ,

$$\mathcal{L}(\alpha_K, \alpha_J^\vee) = 2 \iff K = J.$$

Proof. Write $J = \varepsilon[i, j]$ and $K = \delta[k, l]$. Let

$$E = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{k,j+1}.$$

By Lemma 3, $\mathcal{L}(\alpha_K, \alpha_J^\vee) = \varepsilon\delta E$. The two negative endpoint conditions cannot both hold: if $i = l + 1$ and $k = j + 1$, then $l < i \leq j < k \leq l$, a contradiction. Hence $E \geq -1$. Since also $E \leq 2$, the equality $\varepsilon\delta E = 2$ forces $\varepsilon\delta = 1$ and $E = 2$.

The equality $E = 2$ forces both positive endpoint terms to be 1, so $i = k$ and $j = l$. Since $\varepsilon\delta = 1$ and $\varepsilon, \delta \in \{\pm 1\}$, we also have $\varepsilon = \delta$. Thus $K = J$. The converse is Lemma 4. □

Lemma 6. The maps $\alpha : I_n \times \{\pm 1\} \rightarrow \mathbb{Z}^n$ and $\alpha^\vee : I_n \times \{\pm 1\} \rightarrow (\mathbb{Z}^n)^*$ are injective.

Proof. Suppose first that $\alpha_J = \alpha_K$. Pairing both sides with $\alpha_J^\vee$ gives

$$2 = \mathcal{L}(\alpha_J, \alpha_J^\vee) = \mathcal{L}(\alpha_K, \alpha_J^\vee).$$

Lemma 5 gives $K = J$.

Now suppose that $\alpha_J^\vee = \alpha_K^\vee$. Pairing both sides with α_J gives

$$2 = \mathcal{L}(\alpha_J, \alpha_J^\vee) = \mathcal{L}(\alpha_J, \alpha_K^\vee).$$

Applying Lemma 5 with the roles of the two signed intervals exchanged gives $J = K$. □

We first define $s_J(K)$ for $J, K \in I_n$. Set $L = (J \cup K) \setminus (J \cap K)$. Exactly one of the following cases occurs:

R1. $L = \emptyset$.

R2. $L \in I_n$ and $K \subset J$.

R3. $L \in I_n$ and $K \not\subset J$.

R4. $L \notin I_n \cup \{\emptyset\}$.

Then

$$s_J(K) = \begin{cases} -K & \text{in case R1,} \\ -L & \text{in case R2,} \\ L & \text{in case R3,} \\ K & \text{in case R4.} \end{cases} \quad (2.3)$$

If $J = [i, j]$ and $K = [k, l]$, this is equivalently

$$s_{[i,j]}[k, l] = \begin{cases} -[k, l] & \text{if } (i, j) = (k, l), \\ -[l+1, j] & \text{if } i = k, j > l, \\ [j+1, l] & \text{if } i = k, j < l, \\ -[i, k-1] & \text{if } i < k, j = l, \\ [k, i-1] & \text{if } i > k, j = l, \\ [k, j] & \text{if } i = l+1, \\ [i, l] & \text{if } k = j+1, \\ [k, l] & \text{otherwise.} \end{cases} \quad (2.4)$$

Extend s_J to signed intervals by $s_J(-K) = -s_J(K)$, and extend the first argument by $s_{-J} = s_J$.

Lemma 7. For all $J \in I_n \times \{\pm 1\}$, we have $s_J \in \text{Perm}(I_n \times \{\pm 1\})$.

Proof. A direct case check from (2.4) shows that $s_J(s_J(K)) = K$ for every signed interval K . Hence s_J is its own inverse, and therefore a permutation. $\square$

Lemma 8. For all $J, K \in I_n \times \{\pm 1\}$,

$$\alpha_{s_J(K)} = \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee) \alpha_J.$$

Proof. First assume that $J = [i, j]$ and $K = [k, l]$ are positive intervals. Put $c = \mathcal{L}(\alpha_K, \alpha_J^\vee)$. Lemma 3 gives

$$c = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{k,j+1}.$$

Comparing this value with (2.4) gives:

condition	c	$\alpha_K - c\alpha_J$
$(i, j) = (k, l)$	2	$-\alpha_{[k,l]}$
$i = k, j > l$	1	$-\alpha_{[l+1,j]}$
$i = k, j < l$	1	$\alpha_{[j+1,l]}$
$i < k, j = l$	1	$-\alpha_{[i,k-1]}$
$i > k, j = l$	1	$\alpha_{[k,i-1]}$
$i = l+1$	-1	$\alpha_{[k,j]}$
$k = j+1$	-1	$\alpha_{[i,l]}$
otherwise	0	$\alpha_{[k,l]}$

Each entry in the last column is precisely $\alpha_{s_J(K)}$.

For arbitrary signs, write $J = \varepsilon J_0$ and $K = \delta K_0$ with $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Since $s_{\varepsilon J_0}(\delta K_0) = \delta s_{J_0}(K_0)$ and

$$\mathcal{L}(\alpha_{\delta K_0}, \alpha_{\varepsilon J_0}^\vee) = \delta \varepsilon \mathcal{L}(\alpha_{K_0}, \alpha_{J_0}^\vee),$$

the positive case, multiplied by δ , proves the signed case. $\square$

Lemma 9. For all $J, K \in I_n \times \{\pm 1\}$,

$$\alpha_{s_J(K)}^\vee = \alpha_K^\vee - \mathcal{L}(\alpha_J, \alpha_K^\vee) \alpha_J^\vee.$$

Proof. For positive intervals, symmetry of the type A Cartan matrix gives the same endpoint value for $\mathcal{L}(\alpha_J, \alpha_K^\vee)$ as in the proof of Lemma 8. The case table there used only additivity of sums over adjacent or nested intervals. Replacing each Ae_r by e_r^* in that table gives the desired coroot identity for positive intervals. The signed case follows by the same sign reduction as in Lemma 8. $\square$

Theorem 10. Fix $n \in \mathbb{Z}_{>0}$, and let

$$R = \mathbb{Z}, \quad M = \mathbb{Z}^n, \quad N = (\mathbb{Z}^n)^*.$$

Let $\mathcal{L}$ be the canonical pairing, let $I = I_n \times \{\pm 1\}$ be the set of signed intervals, let α and $\alpha^\vee$ be as in (2.1) and (2.2), and let s be as in (2.3). Then

$$(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$$

is a finite, crystallographic and reduced root pairing.

Proof. The quadruple $(\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, \mathcal{L})$ is the standard perfect pairing between a finite free $\mathbb{Z}$ -module and its dual. It remains to verify the root-pairing axioms from Definition 1.

The maps α and $\alpha^\vee$ are injective by Lemma 6. The identity $\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2$ is Lemma 4. Lemma 7 gives $s_J \in \text{Perm}(I)$ for each J . Finally, Lemmas 8 and 9 give the two reflection formulas. These are exactly the conditions in Definition 1. This proves the tuple is a root pairing. It is finite because I is finite, crystallographic by Lemma 3, and reduced by (2.1) and (2.2). $\square$

2.2 Types B and C

2.3 Type D

2.4 Exceptional types

Chapter 3

Root systems

Definition 11. A root pairing $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ is called a *root system* if the sets $\{\alpha_i : i \in I\}$ and $\{\alpha_i^\vee : i \in I\}$ span the modules M and N , respectively.

Definition 12. Given a root pairing $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$, a set $\Delta \subset I$ is called a *base* of Φ if:

1. the sets $\{\alpha_i : i \in \Delta\}$ and $\{\alpha_i^\vee : i \in \Delta\}$ are R -linearly independent, and
2. for every $x \in I$, there exist $k, m \in \mathbb{Z}_{>0}$, $a_1, \dots, a_k, b_1, \dots, b_m \in \Delta$, and $\varepsilon, \delta \in \{\pm 1\}$ such that

$$\alpha_x = \varepsilon \sum_{r=1}^k \alpha_{a_r} \quad \text{and} \quad \alpha_x^\vee = \delta \sum_{r=1}^m \alpha_{b_r}^\vee.$$

Definition 13. A base Δ is said to be *irreducible* if there exist $k \in \mathbb{Z}_{>0}$ and $a_1, \dots, a_k \in \Delta$ such that:

1. $\{a_1, \dots, a_k\} = \Delta$, and
2. $\mathcal{L}(\alpha_{a_r}, \alpha_{a_{r+1}}^\vee) \neq 0$ for every $1 \leq r < k$.

By definition, a root system admitting an irreducible base is called *irreducible*.

Definition 14. Fix a finite, crystallographic, reduced and irreducible root system $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ and a base $\Delta \subset I$. The pair (Φ, Δ) is said to be of:

1. type *A* if there is a sequence $(i_1, \dots, i_n)$, where $n = |\Delta|$, of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_r}, \alpha_{i_{r+1}}^\vee) = -1 \quad \text{for all } 1 \leq r < n;$$

2. type *B* if there is a sequence $(i_1, \dots, i_n)$ of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_r}, \alpha_{i_{r+1}}^\vee) = -1 \quad \text{for all } 1 \leq r < n-1, \quad \mathcal{L}(\alpha_{i_n}, \alpha_{i_{n-1}}^\vee) = -2;$$

3. type *C* if there is a sequence $(i_1, \dots, i_n)$ of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_r}, \alpha_{i_{r+1}}^\vee) = -1 \quad \text{for all } 1 \leq r < n-1, \quad \mathcal{L}(\alpha_{i_{n-1}}, \alpha_{i_n}^\vee) = -2;$$

4. type D if there exist distinct elements $d_1, d_2, t \in \Delta$ such that

$$\mathcal{L}(\alpha_{d_a}, \alpha_t^\vee) = -1 \quad (a = 1, 2),$$

and $\Delta \setminus \{d_1, d_2\}$ is of type A ;

5. type E if there exist distinct elements $d_1, d_2, t \in \Delta$ such that

$$\mathcal{L}(\alpha_{d_1}, \alpha_{d_2}^\vee) = -1, \quad \mathcal{L}(\alpha_{d_2}, \alpha_t^\vee) = -1,$$

and $\Delta \setminus \{d_1, d_2\}$ is of type A .

We now prove that the root pairings constructed in §2 exhaust all the root system types given in Definition 14. A celebrated theorem of Cartan and Killing states that every finite, crystallographic, reduced and irreducible root system is of exactly one of the types in Definition 14.

Theorem 15. *Set $\Delta = \{[i] : i \in \{1, \dots, n\}\}$. The pair (Φ, Δ) where Φ is the root pairing constructed in Theorem 10 is a root system of type A .*

Proof.

□